

SERIES 3 OF TUTORIAL & LAB TD/TP: ARRAYS

Exercise 1 (Vectors algorithms):

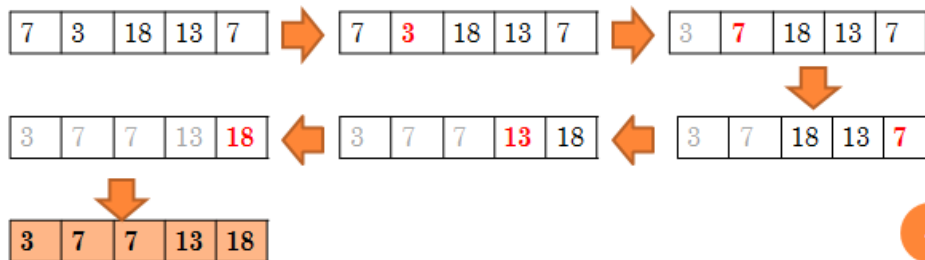
Let A be an integers array of real size N. Write the following algorithms that allow :

1. Read the array elements of A from the keyboard
2. Fill the array A with a given constant.
3. Display the array elements of A
4. Check if an element exists in array A. (sequential search)
5. Count the number of occurrences of a given element in array A.
6. Count the number of positive, negative and zero values in array A.
7. Find the minimum in array A
8. Find the maximum in array A
9. Calculate the largest deviation¹ in array A
10. Test if array A is sorted².

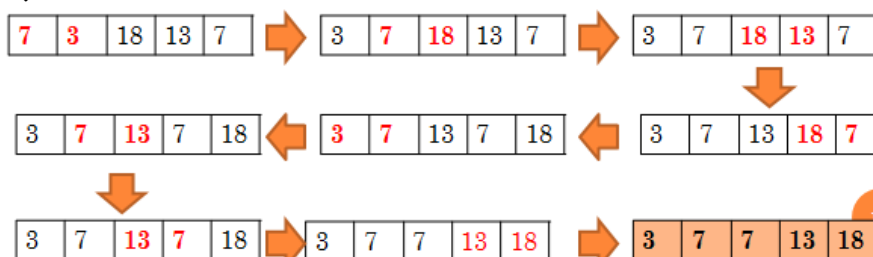
Exercise 2 (Sorting algorithms):

Let A be an integers array of real size N. Write the following sorting algorithms:

1. **Sort by Selection** which consists of searching for the smallest value and placing it at the beginning of the table, then the smallest value in the remaining values and placing it in the second position and so on...



2. **Bubble Sorting** involves progressively moving up the largest elements in a table. The principle is to swap two successive elements if they're not in the right order, and then start again until you can't swap any more.

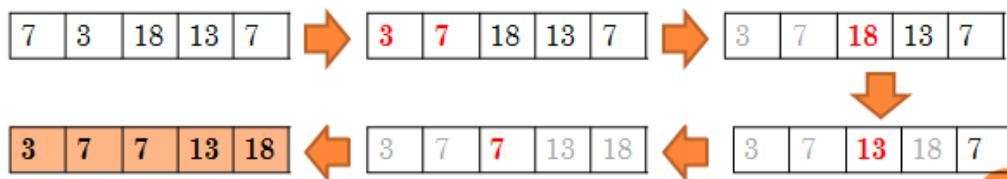


3. **Sort by Insert** which consists of ordering the first two elements and inserting the 3rd element so that the first 3 elements are sorted, then inserting the 4th element in its place and so on. At the

¹The deviation is the absolute value of the difference of two elements

²We speak of an ordered array when the set of possible values is provided with an order relationship. In addition, an array is said to be ordered or sorted in ascending order if any i in $[0, n-1]$: $A[i] \leq A[i+1]$.

end of the i^{th} iteration, the first i elements of array are sorted and arranged at the beginning of array A:



Exercise 3 (Algorithms on sorted vectors):

Let T be an array of ordered integers (sorted in ascending order) of real size N . Write the following algorithms to make:

1. The sequential search for a given item. The principle of this search is to go through the items sequentially, one by one, until the item being searched is found or all (lower) items have been examined.
2. The dichotomous search for a given element. The principle of this search is based on the comparison of the element searched for with the element in the middle of the table. If the element in the middle is equal to the element searched for then the search is finished, otherwise the table is divided into two equal parts: (i) left which contains the elements lower than the element in the middle and (ii) right which contains the elements higher than the element in the middle. We continue the search in the part that may contain the item we are looking for until we find it or arrive at an empty table.
3. Inserting a new item into Table T .
4. Removing an item from Table T .
5. The collation (merging) of two sorted tables so that the resulting table is sorted.

Exercise 4 (Algorithms on string):

Write the following algorithms to:

1. copy a string into another string.
2. copy n characters from one string into another string.
3. concatenate two strings.
4. copy n characters from one string to the end of another string;
5. display the index of the first occurrence of a character in a string.
6. display the index of the last occurrence of the character in a string.
7. convert an integer into a character string.
8. convert a real into a character string.
9. compare the string "ch1" to the string "ch2" and display if ch1 is alphabetically classified before ch2; if ch1 is classified after ch2 and if ch1 = ch2. For example, "Bonjour" is alphabetically before "Hello", so ch1 is smaller than ch2.
10. compare the first n characters of two strings.
11. indicate whether a word (string without space) is a palindrome. A palindrome is a word that reads the same way from right to left and from left to right. Examples: radar, level, madam, civic, ...

12. count the number of words in this string. A word consists of one or more consecutive characters that do not include the white character (space).
13. check if a string is a tautogram or not. A tautogram is a string of words that all start with the same letter.

Exercise 5 (Algorithms on matrices):

Write the algorithms that make it possible to

1. Check if the matrix $A(n,m)$ is Diagonal: if all its elements outside the main diagonal are null.
2. Check if matrix $A(n,m)$ is symmetrical: $a_{ji} = a_{ij}$ for all i and j .
3. Check if matrix $A(n,m)$ is Magic: if the sum of the numbers of each row, each column and each diagonal is equal to the same constant, called magic constant.
4. Construct the transpose of the matrix $A_{n,m} (a_{ij})$ où the rows of A become the columns in $A^t_{m,n} (a_{ji})$
5. Calculate the sum of two matrices $A_{n,m} (a_{ij})$ and $B_{n,m} (b_{ij})$: $s_{ij} = a_{ij} + b_{ij}$
6. Calculate the product of two matrices $A_{n,m} (a_{ij})$ and $B_{m,n} (b_{ij})$ N: $p_{ij} = \sum_{l=0}^{m-1} a_{il} \times b_{lj}$
7. Test if the square matrix $A (n,n)$ is the inverse matrix of $B (n,n)$ (i.e.: $A * B = I$, I identity matrix).

Exercise 6 (Polynomial):

A polynomial of degree n is written with the following form: $a_0 + a_1 * x + a_2 * x^2 + \dots + a_n * x^n$

1. A polynomial can be represented as an integer array. Say how?
2. Using the type in the previous array (question 1), write the algorithms to calculate:
 - a. The value of the polynomial at a given point x .
 - b. The derivative of the polynomial.
 - c. The integral of the polynomial
 - d. The sum of two polynomials.
 - e. The product of two polynomials.
3. A polynomial can be represented as an array of structures. Say how? and repeat the previous operations (from “a” to “e”).